Weak cuts in models of PA

Athar Abdul-Quader

SUNY Purchase

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Strength

Definition (Kirby?)

$I \subseteq_{\text{end}} M$ is strong if whenever $a \in M$, there is $c > I$ such that for all $i \in I$, $(a)_i \in I$ if and only if $(a)_i < c$.

Strong cuts are well-studied; see Kirby's thesis, or Chapter 7 of TSoMoPA. We recall one quick result: recall that $X \subseteq I$ is *coded* if there is $Y \in \text{Def}(M)$ such that $Y \cap I = X$ (or, equivalently, if there is $a \in M$ such that for all $i \in I$, $i \in X$ iff $i \in a$). The collection of all coded sets is denoted $\text{Cod}(M/I)$.

Theorem

Suppose $I \subseteq_{\text{end}} M$ is a cut. Then $(I, \text{Cod}(M/I)) \models \text{ACA}_0$ if and only if I is strong.

Pathologically Defined Cuts

Define $\bigwedge\limits_{i<c} \theta$ inductively so that $\bigwedge\limits_{i<n+1} \theta = (\bigwedge\limits_{i<n} \theta) \wedge \theta$.

Question

Given $\mathcal{M} \models \mathrm{PA}$ countable and recursively saturated, for which cuts $I \subseteq_{\mathrm{end}} M$ is there a truth predicate T such that

$$I = \{c : T(\bigwedge_{i<c}(0=0))\}?$$

- $\omega \subseteq I$
- I is closed under successors.
- What else?

Separability

Definition

I is separable if whenever $a \in M$, there is $c \in M$ such that for all $n \in \omega$, $(a)_n \in I$ if and only if $n \in c$.

Smith (1989): $(\mathcal{M}, T)$ is definably T-saturated:

- Let $(\mathcal{M}, T) \models \mathrm{CT}^-$ and $\langle \phi_i(x) : i < \omega \rangle$ coded.
- Suppose, for each $i \in \omega$ there is $m \in M$ such that $\forall j < i(T(\phi_j(m)))$.
- Then there is $m \in M$ such that for all $i \in \omega$, $T(\phi_i(m))$.

Suppose $I \subseteq_{\text{end}} M$ is such that $I = \{a : (\mathcal{M}, T) \models \bigwedge_{i<a} (0 = 0)\}$. Let $a \in M$, and for each j, define $\phi_j(x)$ as $[\bigwedge_{i<(a)_j} (0 = 0)] \leftrightarrow j \in x$. Smith's result implies the existence of c as in the definition.

Equivalent Definitions

Theorem

Let $\mathcal{M}$ be nonstandard and $I \subseteq_{\mathrm{end}} M$. The following are equivalent:

1. *I is separable: for each a there is c such that whenever $i \in \omega$, $(a)_i \in I$ if and only if $i \in c$.*
2. *There is no a such that $I = \sup(\{(a)_i : i \in \omega\} \cap I) = \inf(\{(a)_i : i \in \omega\} \setminus I)$.*
3. *For each a there is d such that for all $i \in \omega$, $(a)_i \in I$ if and only if $(a)_i < d$.*

(Proof is an exercise.) Notice that (3) is a weakening of *strength*: hence if I is a strong cut, then I is separable.

(2) was used by Jim Schmerl originally in isolating this property in an unpublished note to Enayat (2012). We will use definition (2) often in what follows.

Existence

Notice: if $(\mathcal{M}, I)$ is recursively saturated, then I is separable. (Exercise.) Hence whenever $\mathcal{M}$ is countable and recursively saturated, there are separable cuts I, J, and K such that:

- I is closed under successors but not addition,
- J is closed under addition but not multiplication,
- K is closed under multiplication but not exponentiation, etc

Rationality

Definition (Kossak 1989)

I is rational if for some $a \in M$, either $I \cap \mathrm{Def}(a)$ is cofinal in I and $(M \setminus I) \cap \mathrm{Def}(a)$ is not downward cofinal in $M \setminus I$, or $(M \setminus I) \cap \mathrm{Def}(a)$ is downward cofinal in $M \setminus I$ and $I \cap \mathrm{Def}(a)$ is not cofinal in I.

Looking at the definitions, one can be deluded into thinking that rational cuts are separable. However, the existence of an a such that $\mathrm{Def}(a) \cap I$ is cofinal in I but $\mathrm{Def}(a) \setminus I$ is not downward cofinal in $M \setminus I$ does not ensure that there is no b such that $I = \sup(\{(b)_i : i \in \omega\} \cap I) = \inf(\{(b)_i : i \in \omega\} \setminus I)$. We will need a stronger property to ensure that.

ω-coded

Definition

I is ω-coded if there is $a \in M$ such that either $I = \sup\{(a)_i : i \in \omega\}$ or $I = \inf\{(a)_i : i \in \omega\}$.

If $I = \sup\{(a)_i : i \in \omega\}$, it is ω-coded from below; if $I = \inf\{(a)_i : i \in \omega\}$ it is ω-coded from above. Every rational cut is ω-coded.

To see this: suppose $I \subseteq_{\text{end}} M$ is rational, and a is such that $I \cap \text{Def}(a)$ is cofinal in I and $(M \setminus I) \cap \text{Def}(a)$ is not downward cofinal in $M \setminus I$. Then there is some $c > I$ such that for all terms t, $t(a) \in I$ if and only if $t(a) < c$.

Let t_n be a recursive enumeration of the Skolem terms, and using recursive saturation, find b to code a sequence such that for each $n \in \omega$, $(b)_n = \max(t_n(a), (b)_n + 1)$, if $t_n(a) < c$, or $(b)_n + 1$ otherwise. Then such a b codes an ω-sequence whose supremum is I.

Superrationality

Conversely: are all ω-coded cuts rational? This is true in models in which ω is a *strong cut*. In fact, in that case there is more:

Definition (Kossak 1989)

I is superrational if there is $a \in M$ such that either $I \cap \mathrm{Def}(a)$ is cofinal in I and for all b, $(M \setminus I) \cap \mathrm{Def}(b)$ is not downward cofinal in $M \setminus I$, or $(M \setminus I) \cap \mathrm{Def}(a)$ is downward cofinal in $M \setminus I$ and for all b, $I \cap \mathrm{Def}(b)$ is not cofinal in I.

By replacing Def with Def_k, we get a definition for k-rational and k-superrational. If $\mathcal{M}$ is recursively saturated, then I is 0-superrational if and only if it is superrational.

Clearly every superrational cut is rational, and therefore ω-coded. It turns out: ω is a strong cut if and only if every ω-coded cut is 0-superrational.

Separability vs superrationality

Proposition

For any nonstandard $\mathcal{M} \models \mathrm{PA}$ and $I \subseteq_{\mathrm{end}} M$, the following are equivalent:

1. *I is ω-coded and separable.*
2. *I is 0-superrational.*

To see this: suppose $I = \sup\{(a)_i : i \in \omega\}$. If b is such that $\mathrm{Def}_0(b) \setminus I$ is downward cofinal in $M \setminus I$, then, using bounded satisfaction classes, one can find c such that $(c)_{2n} = (a)_n$ and $(c)_{2n+1} = t_n(b)$ for each $n \in \omega$, where t_n is a recursive enumeration of the Δ_0-definable Skolem functions. This c contradicts one of the equivalent conditions of separability: namely, $I = \sup(\{(c)_i : i \in \omega\} \cap I) = \inf(\{(c)_i : i \in \omega\} \setminus I)$.

Conversely: suppose I is 0-superrational. Then as seen previously, I is ω-coded. It is a routine exercise to see that if $\sup(\{(a)_i : i \in \omega\} \cap I) = I$, then $\inf(\{(a)_i : i \in \omega\} \setminus I) \neq I$.

Separability vs ω-coded

Proposition

For any (nonstandard) $\mathcal{M} \models \mathrm{PA}$:

1. *If ω is a strong cut, then every ω-coded cut is separable.*
2. *If ω is not a strong cut, then every ω-coded cut is not separable.*

(1) $\implies$ (2): next slide. (2) $\implies$ (1): we'll suppose that I is ω-coded so $I = \sup\{(a)_i : i\imath\omega\}$, and assume that the sequence coded by a is increasing. If c is any element coding a sequence of length $> \omega$, notice that $(a)_{(c)_n} \in I$ if and only if $(c)_n \in \omega$. Let b be such that $(b)_n = (a)_{(c)_n}$.

Suppose I is separable. Then there is d such that $(b)_n \in I$ if and only if $n \in d$. That is, $(c)_n \in \omega$ if and only if $n \in d$. Let $p(x)$ be the type $\{n < x : n \in \omega\} \cup \{n \in d \leftrightarrow (c)_n < x : n \in \omega\}$. Any realization of this type is a witness to strength (for the sequence coded by c).

Existence?

The following are equivalent for any $\mathcal{M}$ (Kossak 1989, Theorem 6.5):

1. ω is a strong cut in $\mathcal{M}$.
2. ω is 0-superrational in $\mathcal{M}$.
3. For every $I \subseteq_{\text{end}} M$, I is coded by ω if and only if I is 0-superrational in $\mathcal{M}$.
4. There exists a 0-superrational cut I in $\mathcal{M}$.

Arithmetic Properties

If $I \subseteq_{\text{end}} M$, then $I \models \mathrm{I}\Sigma_0$.* If I is separable, rational, superrational, can we get anything more?

Separable: no, since we know there are separable cuts not closed under addition, separable cuts closed under addition but not multiplication, etc.

These cuts were I such that $(\mathcal{M}, I)$ is recursively saturated. Such I are **never** ω-coded, and therefore not rational. Are rational cuts necessarily closed under addition, multiplication, etc? No!

- For any a, b, if $b \notin \mathrm{Def}(a)$, then $\sup\{t(a) : t(a) < b\}$ is a rational cut.
- Pick any a, use recursive saturation to find b such that $a < b < 2a$ and $b \notin \mathrm{Def}(a)$.

(Similarly: can pick b such that $n \cdot a < b < a^2$)

*Need to be careful since I might not be closed under addition.

Summarizing

In countable, recursively saturated $\mathcal{M}$:

- Superrational $\implies$ rational $\implies$ ω-coded.
- If ω is strong, then superrational = rational = ω-coded and separable.
- In every nonstandard $\mathcal{M}$, there exist separable cuts that are not rational.
- If ω is not strong, then there are no superrational cuts, and separable cuts are never ω-coded (and therefore not rational).
- Regardless: there always exist rational cuts (example on previous slide).
- None of these properties (separable, rational, superrational) imply any amount of arithmetic.

Automorphisms

Definition (Kossak 1989)

I is almost rigid if for some $a \in M$, either

$$I = \sup(\{t(a) : t \in \text{Term}\} \cap I)$$

or

$$I = \inf(\{t(a) : t \in \text{Term}\} \setminus I).$$

The name comes from the fact that if $\mathcal{M}$ is countable and recursively saturated, then $I \subseteq_{\text{end}} M$ is almost rigid if and only if it has at most countably many automorphic images. Clearly form the definition, rational cuts are almost rigid.

Semiregularity

In his PhD thesis (1977), Kirby made an analogy between properties of cuts in models of PA and (large) cardinal properties in set theory. We will briefly review them here to contrast them with separable / rational / superrational. We already defined *strong* cuts; next we will define semiregular and regular cuts.

Definition

$I \subseteq_{\text{end}} M$ is semiregular if whenever f is a coded function in $\mathcal{M}$ with domain bounded in I, $\text{rg}(f) \cap I$ is bounded in I.

The set-theoretic inspiration is more direct when we use the notion of cofinality. $\text{cf}^{\mathcal{M}}(I)$ is defined to be the intersection of all $J \subseteq_{\text{end}} M$ such that there is $f \in M$ with $\sup(f(J)) = I$.

Proposition

$I \subseteq_{\text{end}} M$ is semiregular if and only if $\text{cf}^{\mathcal{M}}(I) = I$.

Compare to set theory: κ is regular if $\text{cf}(\kappa) = \kappa$.

Regularity

Definition

$I \subseteq_{\text{end}} M$ is regular if for every $f \in M$, whenever $I \subseteq \text{dom}(f)$ and $\text{rg}(f \restriction I)$ is bounded in I, then there is $i \in I$ such that $f^{-1}(i) \cap I$ is unbounded in I.

Regular cuts are those cuts $I \subseteq_{\text{end}} M$ such that there exist $\mathcal{M} \prec_{\text{cof}} \mathcal{N}$ with $\text{GCIS}(\mathcal{N}, \mathcal{M}) = I^{\dagger}$ and $c \in N \setminus M$ such that $I < c < M \setminus I$.

- Exercise: strong cuts are regular.
- Exercise: regular cuts are semiregular.
- Harder (but in Kirby's thesis): there are semiregular cuts that are not regular.
- (TSoMoPA Theorem 7.2.2): Every countable $\mathcal{M}$ has an elementary end extension in which it is semiregular but not regular.

†GCIS = Greatest Common Initial Segment

Unlike separable / rational / superrational cuts: semiregular / regular / strong cuts are all closed under addition, multiplication, exponentiation, much more:

- $I \subseteq_{\text{end}} M$ semiregular $\implies I \models \mathrm{I}\Sigma_1$.
- $I \subseteq_{\text{end}} M$ regular $\implies I \models \mathrm{B}\Sigma_2$.
- $I \subseteq_{\text{end}} M$ strong $\implies I \models \mathrm{PA}$.

In fact, one can characterize semiregular, regular, and strong cuts in terms of the second order properties of $(I, \mathrm{Cod}(M/I))$:

- $I \subseteq_{\text{end}} M$ semiregular $\Leftrightarrow (I, \mathrm{Cod}(M/I)) \models \mathrm{I}\Sigma_1$.
- $I \subseteq_{\text{end}} M$ regular $\Leftrightarrow (I, \mathrm{Cod}(M/I)) \models \mathrm{B}\Sigma_2$.
- $I \subseteq_{\text{end}} M$ strong $\Leftrightarrow (I, \mathrm{Cod}(M/I)) \models \mathrm{ACA}_0$.

Clearly, then: strong $\implies$ regular $\implies$ semiregular.

Implications and Non-implications

Clearly strong cuts are separable. Is there anything else we can say about strong / regular / semiregular cuts vs (super)rational / separable cuts?

- Notice: if I is ω-coded, then I is not semiregular.
- If I is rational or superrational, then it is ω-coded. Hence it is not semiregular.
- What about separability? If I is not separable, then the a which witnesses $I = \sup(\{(a)_i : i \in \omega\} \cap I)$ immediately witnesses that I is not semiregular.
- Hence: Semiregular $\implies$ separable!

Thank you!

Some references:

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